Topic: Cartesian Product Operation

Q. What is a Cartesian Product in DBMS? Explain its working mechanism and provide an example demonstrating how it combines two relations.

> 📌 **Definition to Remember**
> The **Cartesian Product** (also called Cross Product, denoted ×) is a **binary operation** in relational algebra that pairs **every row of the first relation with every row of the second relation**, producing all possible combinations. Output columns = sum of both columns; Output rows = product of both row counts. Union Compatibility is **NOT** required.

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1. Cartesian Product Operator (×)

- **Binary operation** — operates on two tables (which need NOT be union compatible).
- **Syntax:** Relation_A × Relation_B
- **Output Degree (Columns):** Columns_A + Columns_B
- **Output Cardinality (Rows):** Rows_A × Rows_B

2. Working of Cartesian Product

- Takes **every single row from A** and pairs it with **every single row from B**.
- Generates all possible row combinations — most of which are **meaningless on their own**.
- On its own, the Cartesian Product produces massive redundant data.
- It is primarily used as the **first step for JOIN operations**: Apply Cartesian Product → Apply Selection (σ) to filter meaningful rows.

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A × B :
Row1_A  ↔  Row1_B
Row1_A  ↔  Row2_B
Row2_A  ↔  Row1_B
Row2_A  ↔  Row2_B
(2 × 2 = 4 combinations)
```

3. Example

COLORS (2 rows)

Color_ID	Color_Name
1	Red
2	Blue

SHAPES (2 rows)

Shape_ID	Shape_Name
A	Circle
B	Square

Query: COLORS × SHAPES
Output: 2 + 2 = **4 columns**; 2 × 2 = **4 rows**

Color_ID	Color_Name	Shape_ID	Shape_Name
1	Red	A	Circle
1	Red	B	Square
2	Blue	A	Circle
2	Blue	B	Square

⭐ **Must-Write Points (for 10 marks)**

1. Cartesian Product (×) = Cross Product — pairs every row of A with every row of B.
2. Output columns = Columns_A + Columns_B; Output rows = Rows_A × Rows_B.
3. Union Compatibility is **NOT** required — the two relations can be completely different.
4. On its own, produces large amounts of **meaningless data**.
5. It is the foundation of **JOIN operations**: Cartesian Product + Selection = Join.
6. If A has m rows and B has n rows, the result has m × n rows.
7. Cartesian Product is a basic (fundamental) operator in relational algebra.

⚡ **Quick Recall**

`× (Cartesian Product) → Binary → All Row Combinations → Cols = A+B, Rows = A×B → No Union Compatibility Needed → Foundation of JOIN`